

THESIS II PRESENTATION

Controllable Motion-Guided Image Editing

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Reference-time control of Stable Diffusion using differentiable RAFT

Stable Diffusion

RAFT

DDIM

ML

Agenda

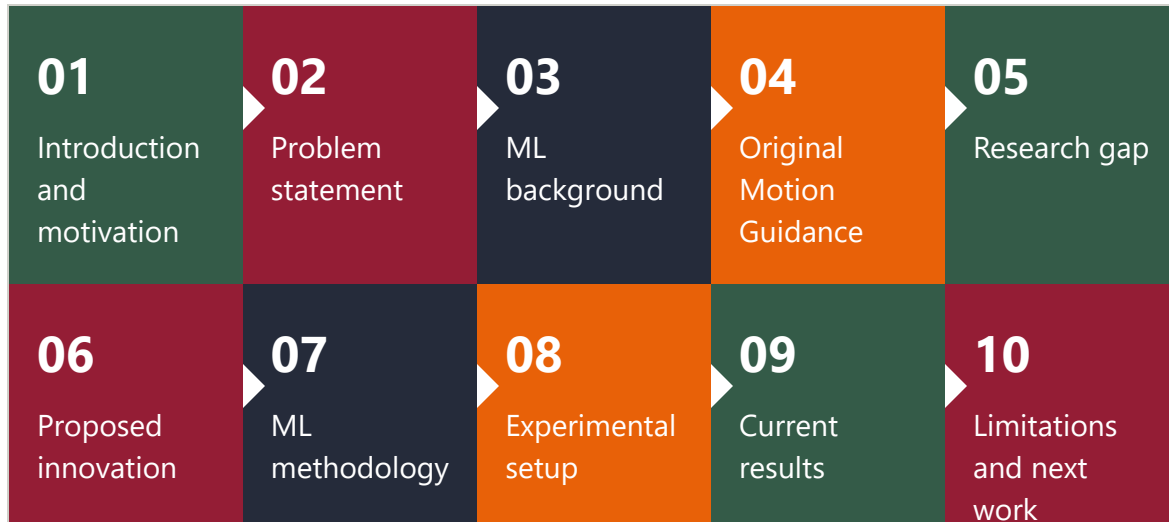


IMAGE EDITING

Generative models create realistic images, but exact spatial control is difficult

Text is semantic

Prompts describe what should appear,

Editing needs geometry

Object motion requires location,

Scenes must remain stable

The moved object,

Why controllable motion editing matters

01

Creative design

Move or reshape objects

02

Visual prototyping

Test alternative compositions from one

03

Human control

Replace vague prompt instructions

04

ML research

Study how external

How can a pretrained latent diffusion model follow explicit object motion without retraining?

- Original Motion Guidance depends on prepared dense flow fields.
- Soft guidance can be weak for large displacement.

Control = target
motion + learned
image prior +
source

Build an inference-time ML control pipeline



Represent motion

Generate dense target flow from simple primitives.



Guide diffusion

Use differentiable RAFT losses during DDIM



Preserve content

Protect non-edited latent regions and



Repair background

Handle the source hole after object

Pretrained models used in this thesis

AutoencoderKL

Encodes RGB images into four-channel latent tensors and decodes predictions.

CLIP ViT-L/14

Transforms the text prompt into a frozen conditioning representation.

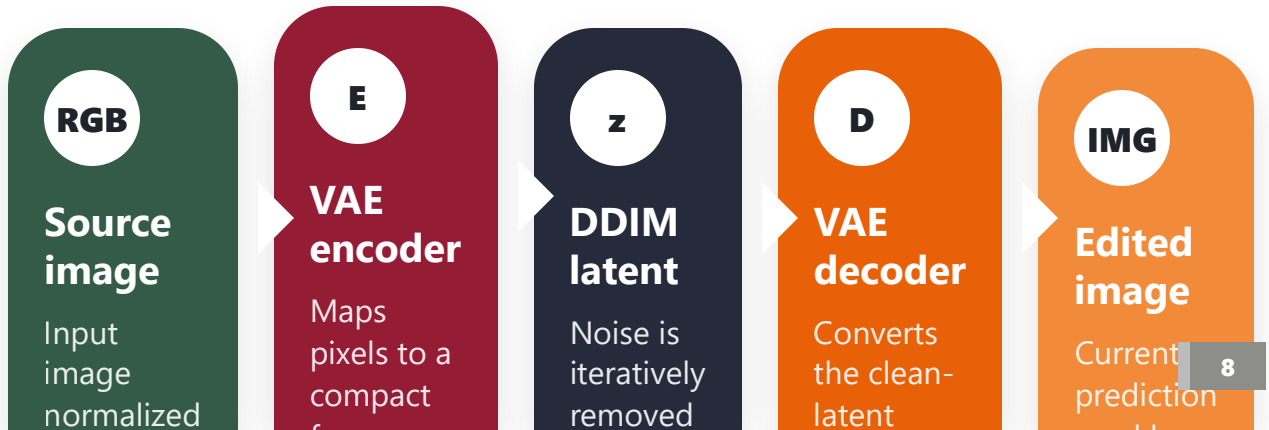
Stable Diffusion U-Net

Predicts noise under timestep and cross-attention

RAFT

Predicts dense optical flow and provides differentiable motion supervision

Stable Diffusion works in compressed latent space



CLIP and classifier-free guidance retain semantic control

- CLIP encodes the prompt once before sampling.
- The U-Net predicts conditional and unconditional noise.
- Classifier-free guidance combines both predictions.

$$\epsilon_{cfg} = \epsilon_u + s(\epsilon_c - \epsilon_u)$$

RAFT converts visual change into dense motion

RAFT input

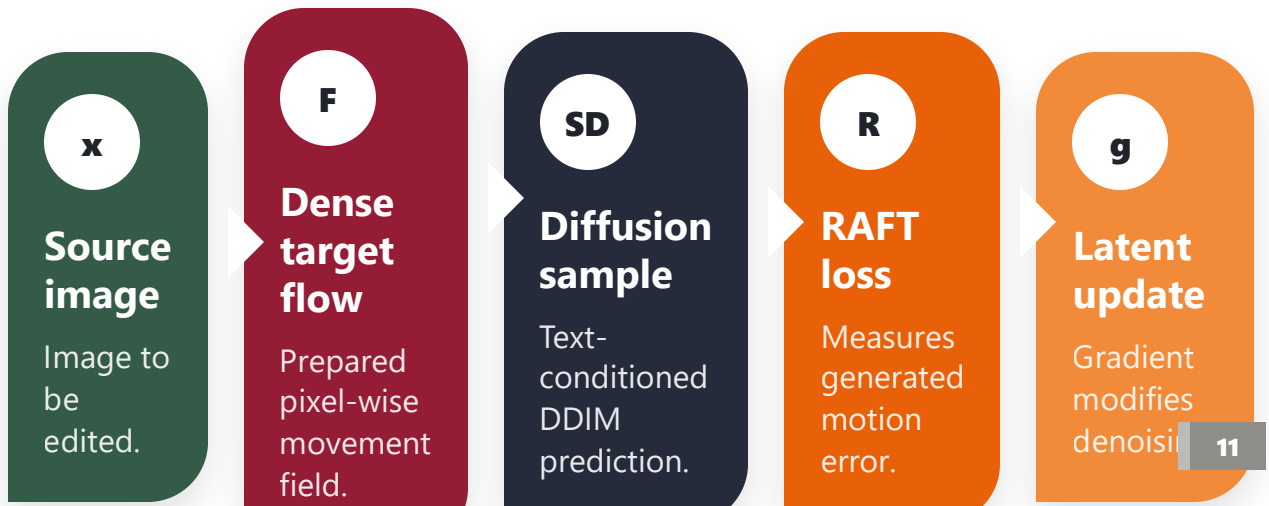
Source image and current decoded diffusion prediction.

$$\hat{F} = \text{RAFT}(x_{\text{source}}, \dots)$$

RAFT role

- Estimates displacement for every pixel.
- Compares generated motion with target motion.

Original Motion Guidance



What is missing in the original workflow?

Dense-flow dependency

Users need a prepared target flow instead of simple edit parameters.

Weak large motion

Soft latent guidance may not enforce a large rigid

Global influence

External gradients can affect areas outside the intended

Disocclusion

The original object location needs plausible background reconstruction.

How my work differs from Motion Guidance

Original Motion Guidance

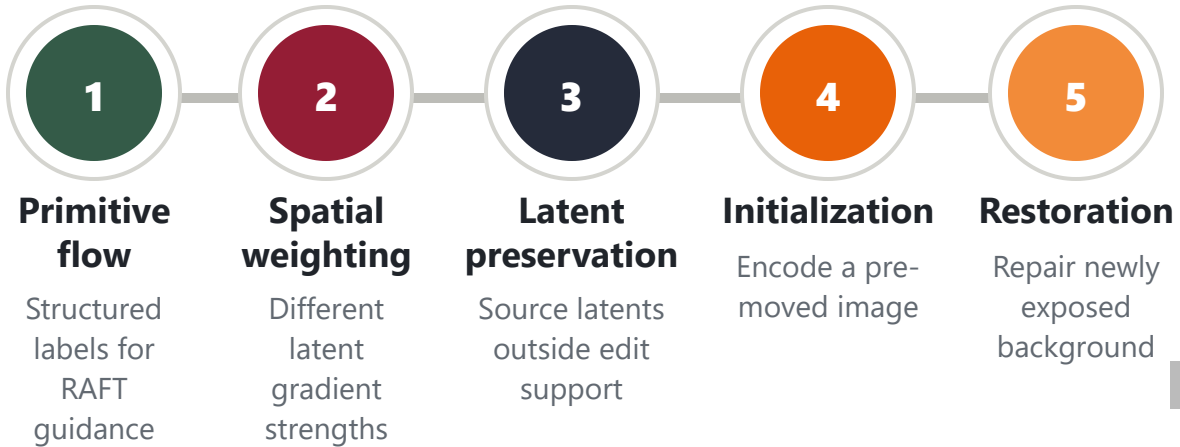
- Precomputed dense flow input
- Global external guidance

VS

Proposed extension

- Primitive-generated target supervision
- Mask-weighted

Five additions around the pretrained ML models



Simple motion parameters become dense RAFT supervision

Translation

Horizontal and vertical offsets: d_x , d_y .

Scale

Uniform object expansion or contraction around its center.

Stretch

Different horizontal and vertical scale factors.

Rotation

Pixel displacement induced by an angle around the mask center

Two losses control motion and appearance

Flow loss

Matches RAFT-predicted motion with the target field.

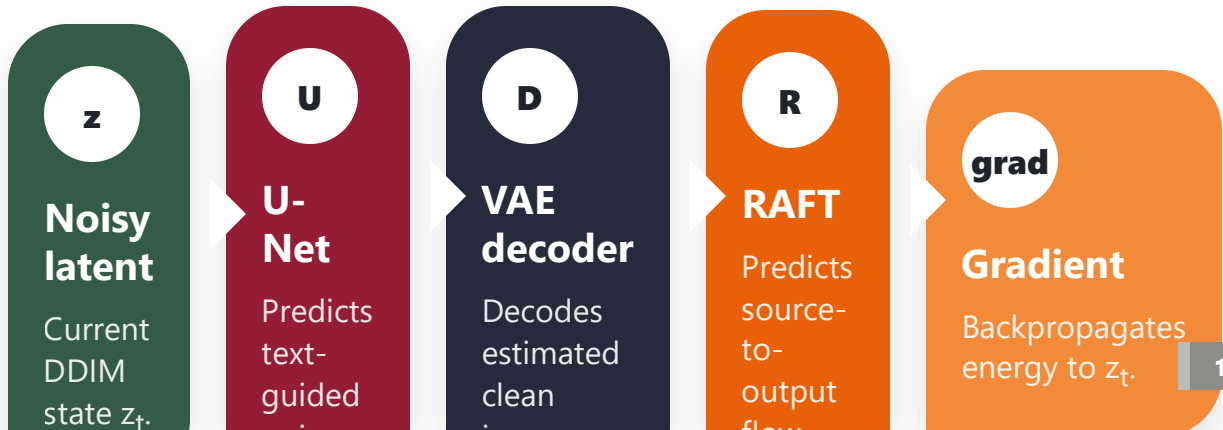
$$L_{flow} = \text{MAE}(\hat{F}, F)$$

Color loss

Warps the current prediction and compares it with the source.

$$L_{color} = \text{MAE}(x_0, W(x, F_{warp}))$$

RAFT supervises Stable Diffusion through backpropagation



Mask-dependent guidance localizes the ML update

- Stronger gradient weight inside the editable support.
- Lower gradient weight outside the object region.
- Applied directly to the latent guidance tensor.

$$\tilde{g}_t(p) = w(p)g_t(p)$$

$$w_{in} > w_{out}$$

Protect unedited regions during stochastic denoising

Cached inversion latents

Reuse source-consistent latent states when available.

Forward-noised source

Fallback when cached latent steps are missing.

RePaint-style replacement

Copy source latents outside the editable region at each step.

Large displacement starts from a stronger latent state



Move object

Construct a geometrically shifted image.



Encode

Use the same Stable Diffusion VAE.



Add noise

Match the initial DDIM noise level.



Refine

Run U-Net denoising with RAFT guidance.

Modified denoising update

- Compute classifier-free noise prediction.
- Decode the estimated clean latent.
- Evaluate RAFT flow and color energy.
- Backpropagate, spatially weight, and clip the gradient.

$$\varepsilon' = \varepsilon_{cfg} - \sqrt{(1 - \alpha_t)} \eta_t \tilde{g}_t$$

Controls for stable inference

Gradient clipping

Limits excessive latent updates caused by large loss gradients.

Guidance schedule

Guidance is disabled during the final 20% of DDIM steps.

Recursive steps

Supports repeated denoising at a timestep when configured.

RAFT iterations

Controls recurrent flow refinement and computational cost.

Motion creates a second problem: missing background

- The original object location becomes disoccluded.
- RAFT guidance describes motion, not the hidden background.
- Optional LaMa adds learned large-mask inpainting

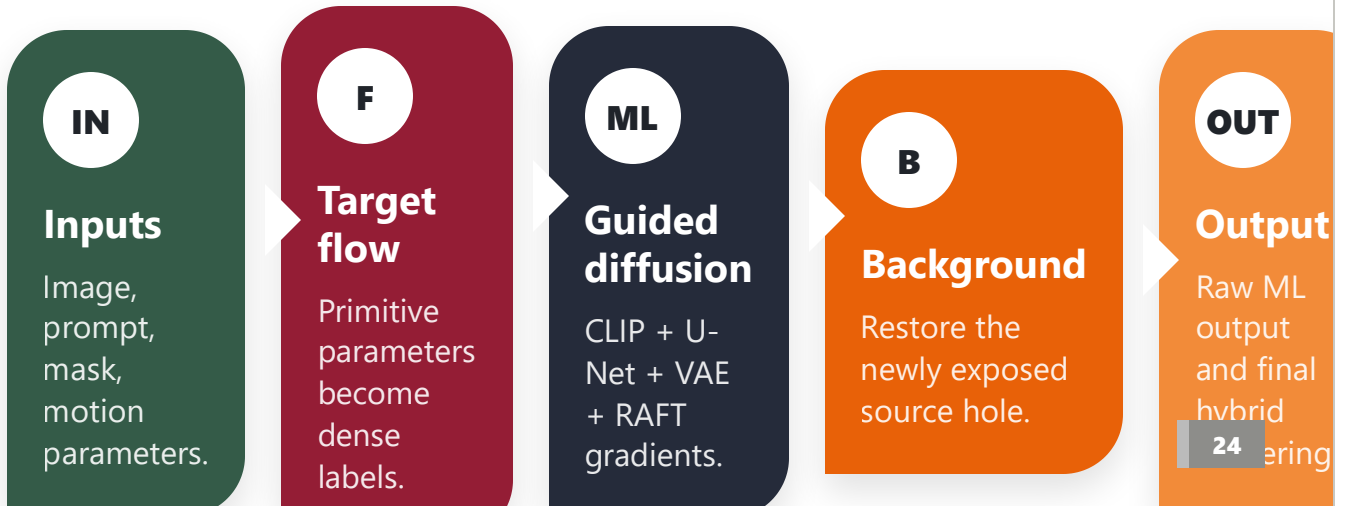
Important separation

Motion guidance: where content should move.

Restoration: what should appear behind it.

Compositing: how the sharp source object is placed in the final contained-translation

Complete inference workflow



Documented inference settings

120

DDIM steps

Iterative latent denoising.

6.5

CFG scale

Text-conditioning strength.

30

Guidance weight

External motion-gradient scale.

60

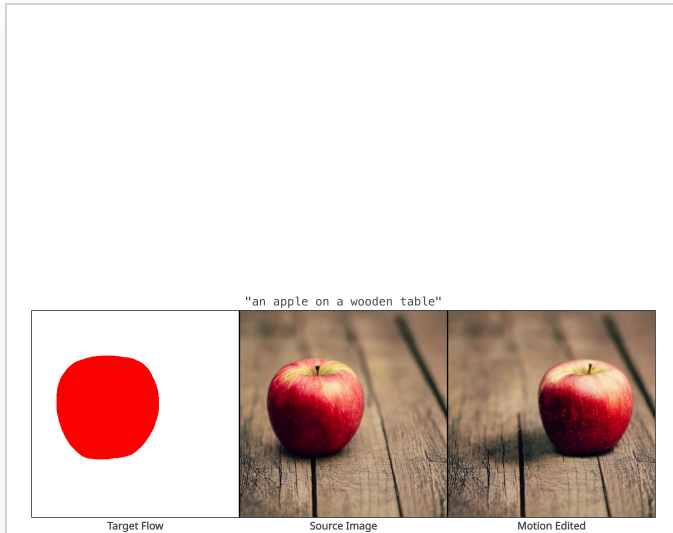
Gradient clip

Maximum update norm.

What the sampler records

Total energy	Combined flow and color guidance objective	Trend
Flow loss	Difference between RAFT prediction and target flow	Lower
Color loss	Appearance mismatch after differentiable warping	Lower
Noise norm	Magnitude of the U-Net noise prediction	Stable

Contained rigid translation



- Explicit rightward displacement is achieved.
- Source apple texture is preserved.
- Old location is filled with table texture.
- Final saved image is a

Different components solve different problems

Primitive flow

Makes motion supervision easy to specify.

RAFT guidance

Measures whether generated motion follows the target.

Initialization

Enforces a strong starting displacement for rigid motion.

Restoration

Completes background that motion guidance cannot observe

Current limitations

- Only one directly inspectable result.
- No completed quantitative comparison.
- No fixed-seed controlled ablation.
- Final composite can hide the

Immediate improvements

- Save raw diffusion and all intermediate stages.
- Add reproducible seeds and automatic metrics.
- Plot saved loss and gradient trajectories.
- Test multiple images

MAIN TAKEAWAY

Pretrained generative and motion models can interact through inference-time gradients

Thank you

Questions and feedback

My contribution is an extension of Motion Guidance primitive